

Cellphone-Based Remote Controller for Water Pump



ABDUL NASIR K.T.

Inconvenience in switching on a water pump installed in a remote farm is a common problem faced by farmers. Many circuits have been developed to solve this problem. Most of them are expensive and micro-controller-based. Here we present a cellphone-based remote controller for water pump. By calling the cellphone attached to the controller, the water pump can be directly activated.

Circuit and working

Fig. 1 shows the block diagram of cellphone-based remote controller for water pump. Fig. 2 shows the circuit. The circuit is built around DTMF decoder IC MT8870 (IC1), timer NE555 (IC2) wired as monostable multivibrator and a few discrete components. The main component of the circuit is IC MT8870. This DTMF decoder has

band-split filter and digital decoder functions. It offers the advantages of small size, low power consumption and high performance.

Once monostable timer IC2 is triggered, its output goes high for the preset time period. The time period depends on the values of resistor R7 and capacitor C4. It can be adjusted between 8 and 50 minutes using potmeter VR1. The high output at pin 3 of IC2 energises relay RL1 to switch on the water pump.

The triggering pulse for IC2 is generated by DTMF decoder IC1 and the arrangement of diodes D1 through D5.

Test Points

Test point	Details
TP0	GND
TP1	5V (when S1 on)
TP2	Triggering pulse for timer IC2
TP3	5V (water pump enabled)

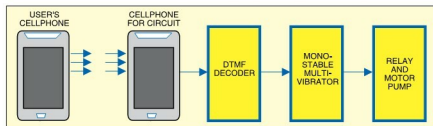


Fig. 1: Block diagram of cellphone-based remote controller for water pump

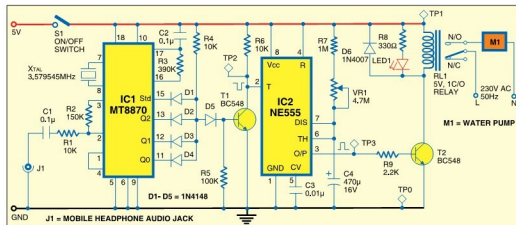


Fig. 2: Circuit of cellphone-based remote controller for water pump

PARTS LIST

Semiconductors:

- IC1 - MT8870 DTMF decoder
- IC2 - NE555 timer
- T1-T2 - BC548 transistor
- D1-D5 - 1N4148 diode
- D6 - 1N4007 rectifier diode
- LED1 - 5mm red LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R4, R6 - 10-kilo-ohm
- R2 - 150-kilo-ohm
- R3 - 390-kilo-ohm
- R5 - 100-kilo-ohm
- R7 - 1-mega-ohm
- R8 - 330-ohm
- R9 - 2.2-kilo-ohm
- VR1 - 4.7-mega-ohm potmeter

Capacitors:

- C1, C2 - 0.1 μ F ceramic disk
- C3 - 0.01 μ F ceramic disk
- C4 - 470 μ F, 16V electrolytic

Miscellaneous:

- M1 - Water pump
- S1 - On/off switch
- RL1 - 5V, 1C/O relay
- J1 - Audio jack
- X₁AL - 3.579545MHz crystal

Std pin of IC1 provides a high pulse when a valid tone-pair is received. Transistor T1 conducts only when outputs Q0 through Q2 and Std are high simultaneously. This can be achieved by sending digit '7' through DTMF.

The water pump controller is connected to a dedicated cellphone through connector J1 with auto-answering mode enabled. The DTMF signal sent from the user end is decoded by the DTMF decoder and the corresponding binary-coded decimal (BCD) value appears on outputs Q0 through Q3. In this circuit only three of them are used.

Working of the circuit is simple. To switch 'on' the water pump, call the cellphone connected to the controller circuit and press '7' once the ring stops. LED1 will glow to indicate that the water pump is switched on. The water pump turns off automatically after the preset time. LED1 turns off simultaneously.